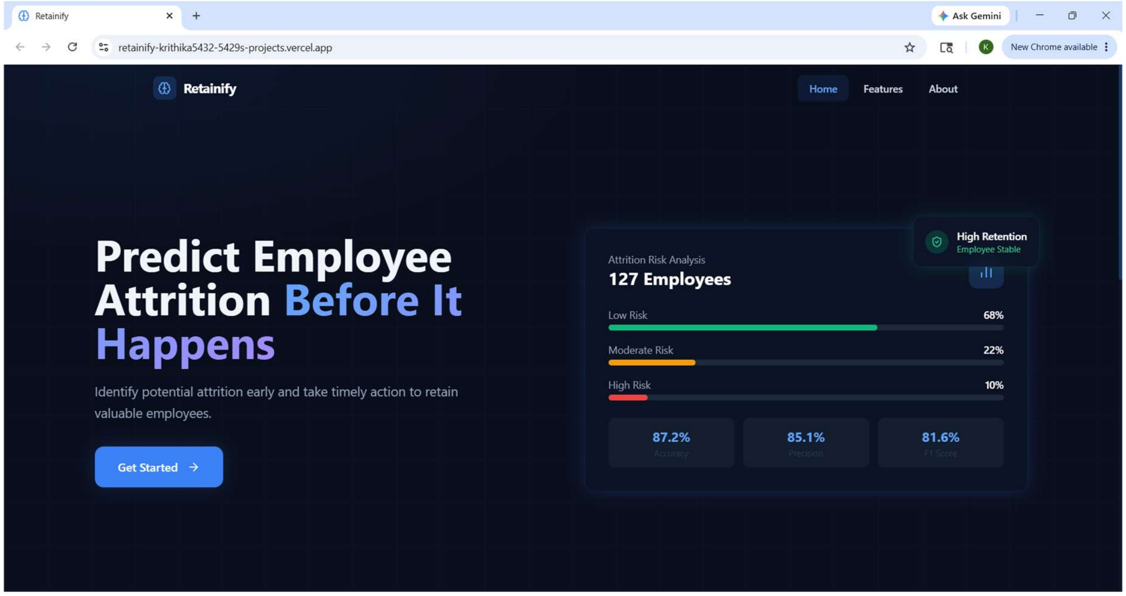
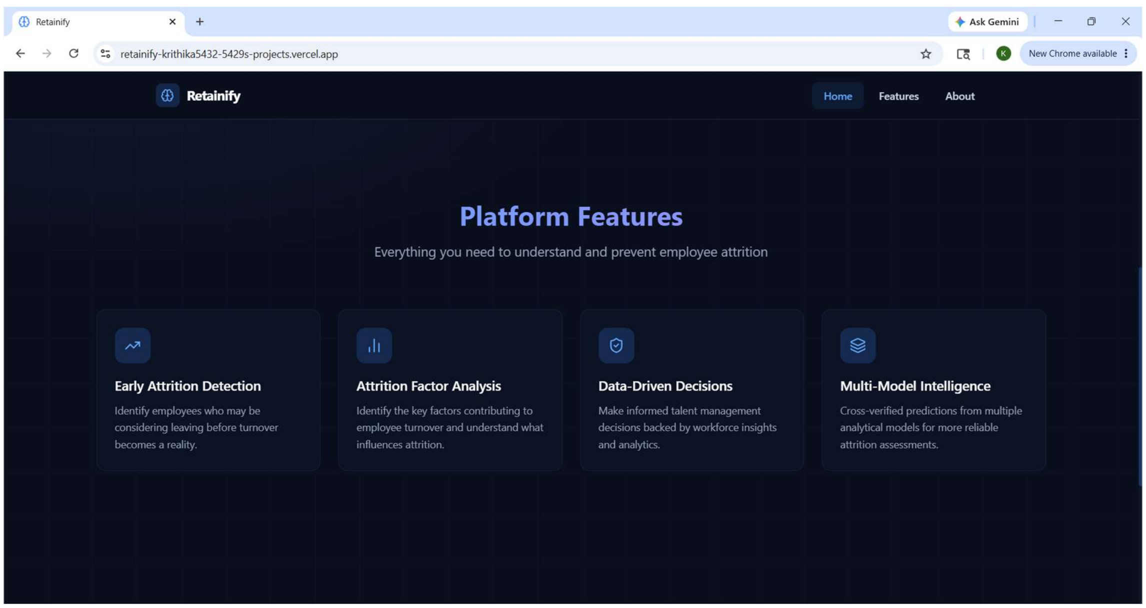
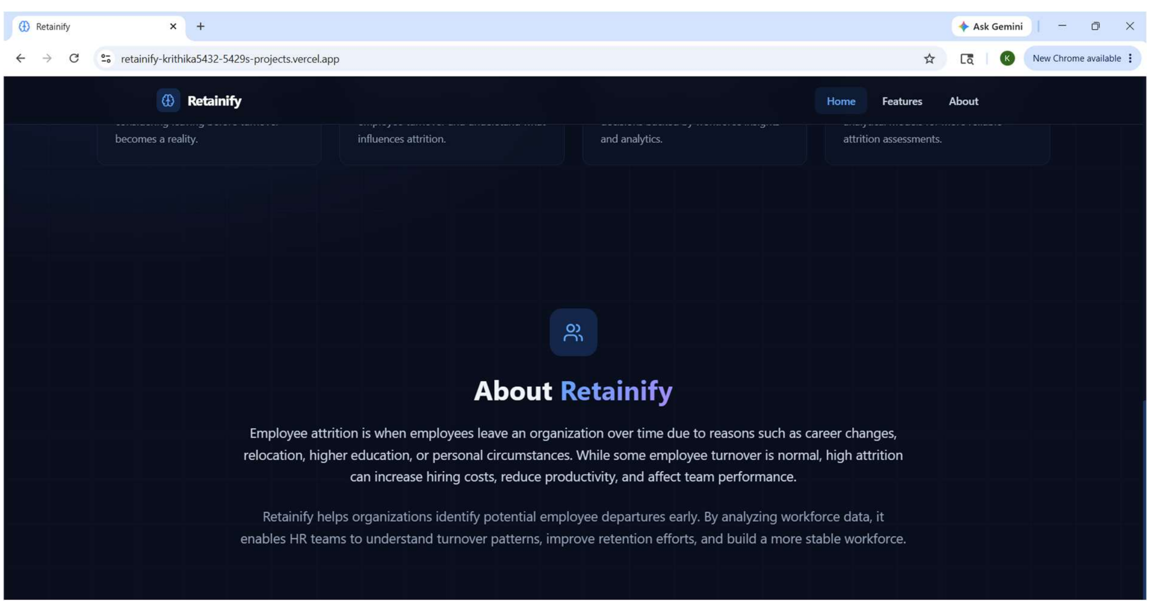
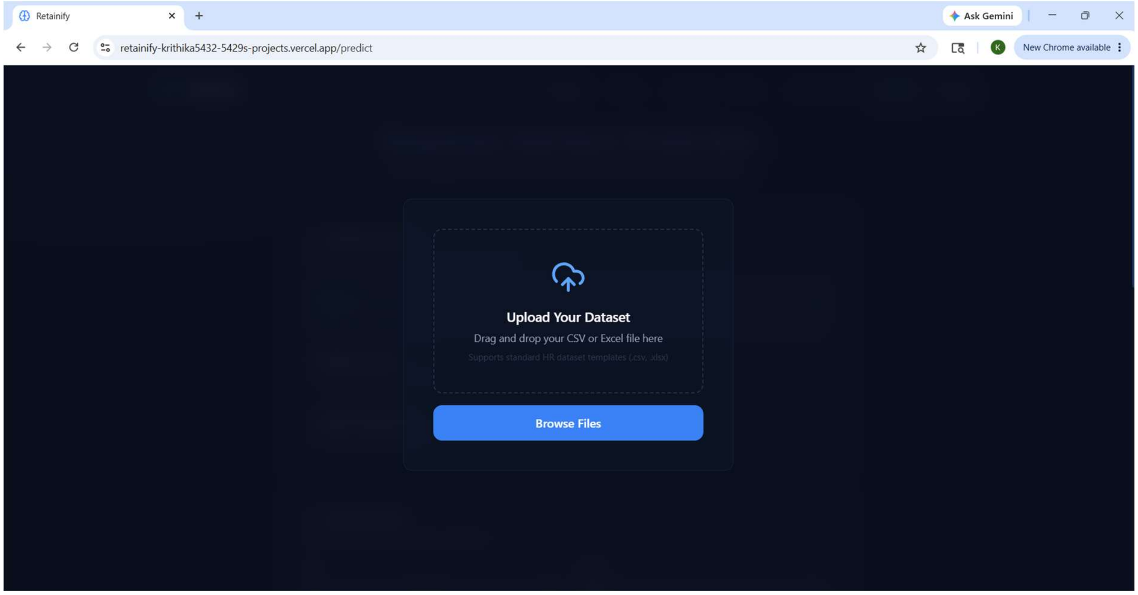
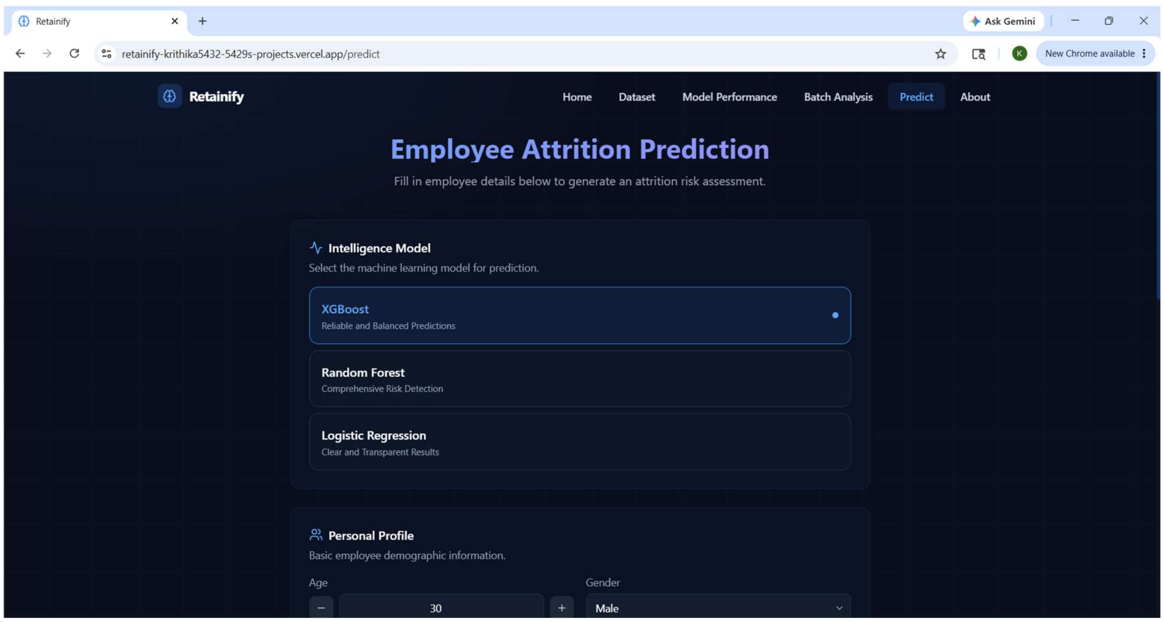
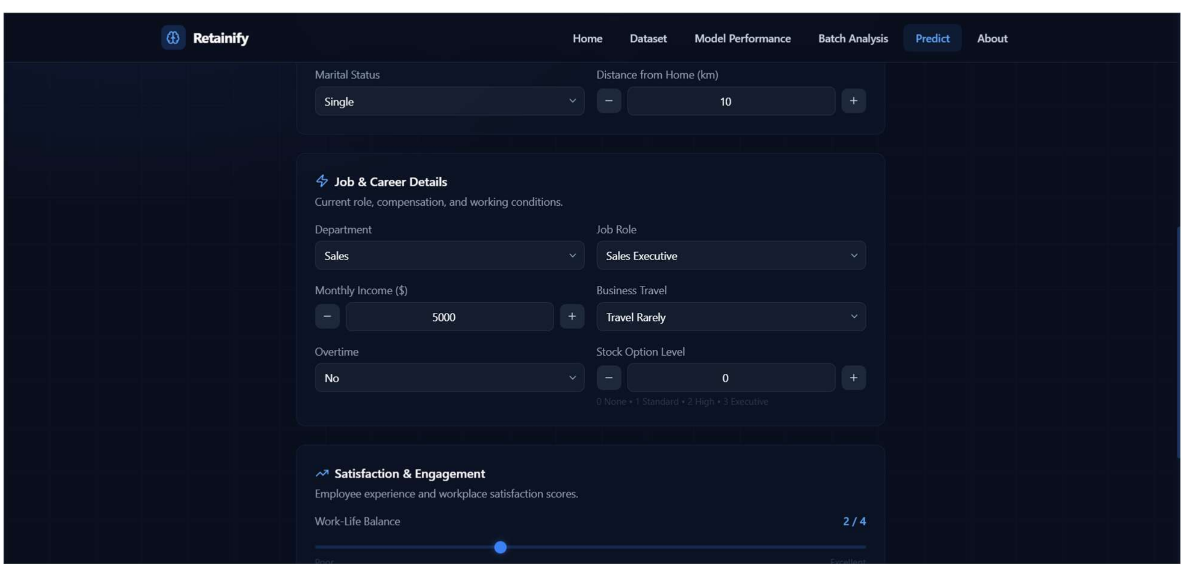

Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app
New Chrome available
Retainify
Home
Features
About
Predict Employee Attrition Before It Happens
Identify potential attrition early and take timely action to retain valuable employees.
Get Started
High Retention
Employee Stable
Attrition Risk Analysis
127 Employees
Low Risk
68%
Moderate Risk
22%
High Risk
10%
87.2%
85.1%
81.6%

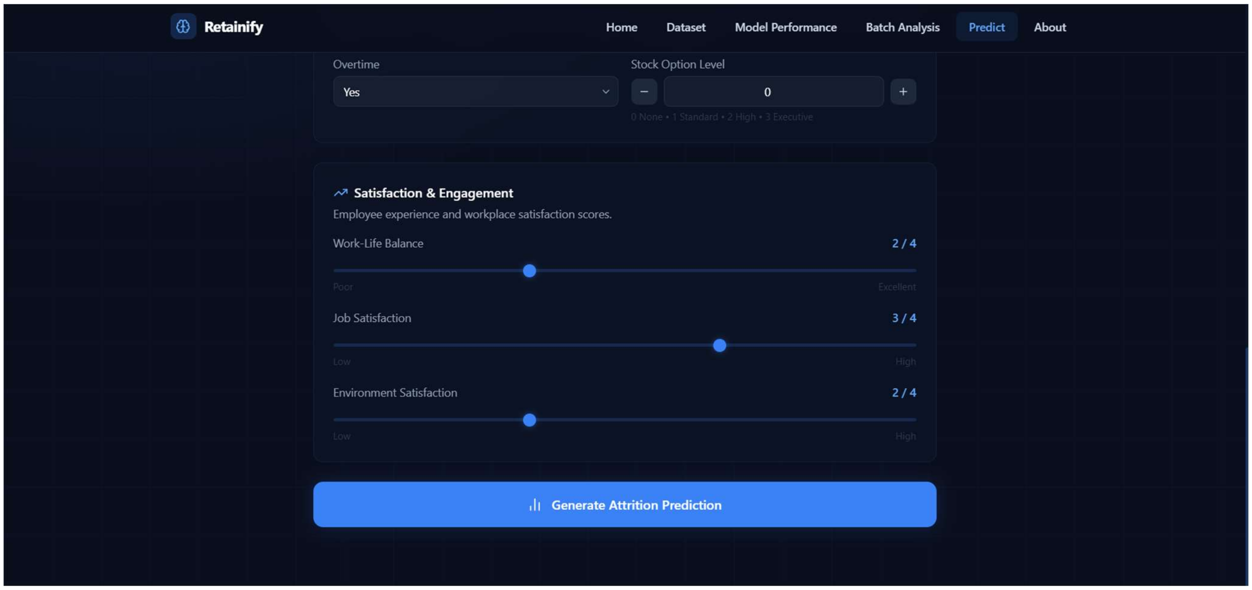

Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app
New Chrome available
Retainify
Home
Features
About
Platform Features
Everything you need to understand and prevent employee attrition
Early Attrition Detection
Identify employees who may be considering leaving before turnover becomes a reality.
Attrition Factor Analysis
Identify the key factors contributing to employee turnover and understand what influences attrition.
Data-Driven Decisions
Make informed talent management decisions backed by workforce insights and analytics.
Multi-Model Intelligence
Cross-verified predictions from multiple analytical models for more reliable attrition assessments.

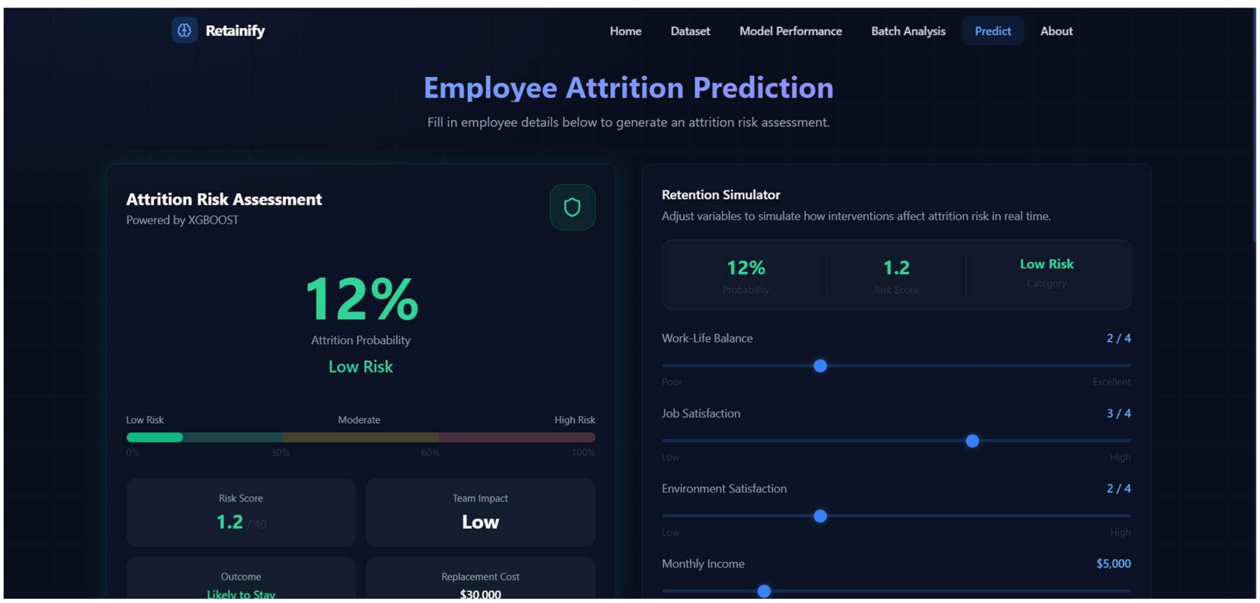

Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app
New Chrome available
Retainify
Home
Features
About
becomes a reality.
influences attrition.
and analytics.
attrition assessments.
About Retainify
Employee attrition is when employees leave an organization over time due to reasons such as career changes, relocation, higher education, or personal circumstances. While some employee turnover is normal, high attrition can increase hiring costs, reduce productivity, and affect team performance.
Retainify helps organizations identify potential employee departures early. By analyzing workforce data, it enables HR teams to understand turnover patterns, improve retention efforts, and build a more stable workforce.

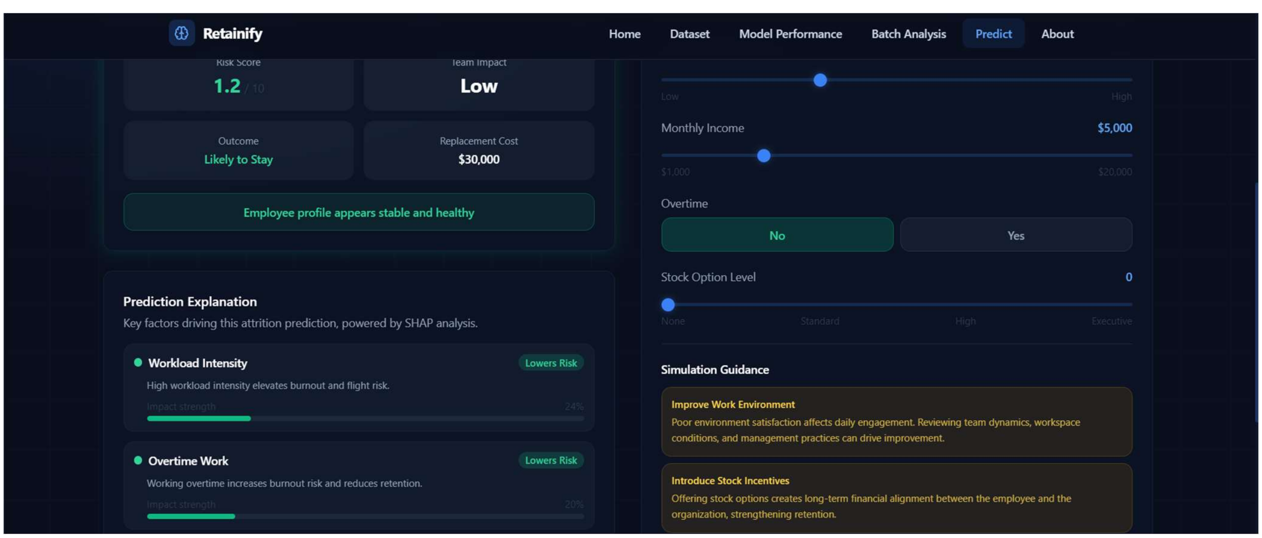

Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app/predict
New Chrome available
Upload Your Dataset
Drag and drop your CSV or Excel file here
Browse Files

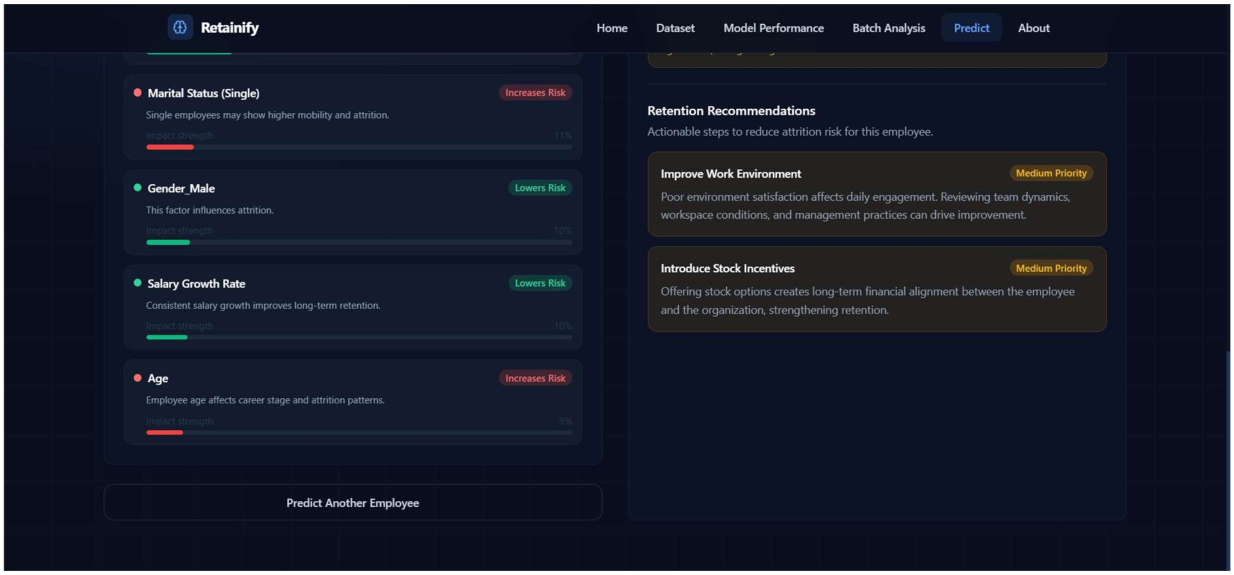

Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app/predict
New Chrome available
Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Employee Attrition Prediction
Fill in employee details below to generate an attrition risk assessment.
Intelligence Model
Select the machine learning model for prediction.
XGBoost
Reliable and Balanced Predictions
Random Forest
Comprehensive Risk Detection
Logistic Regression
Clear and Transparent Results
Personal Profile
Basic employee demographic information.
Age
30
Gender
Male

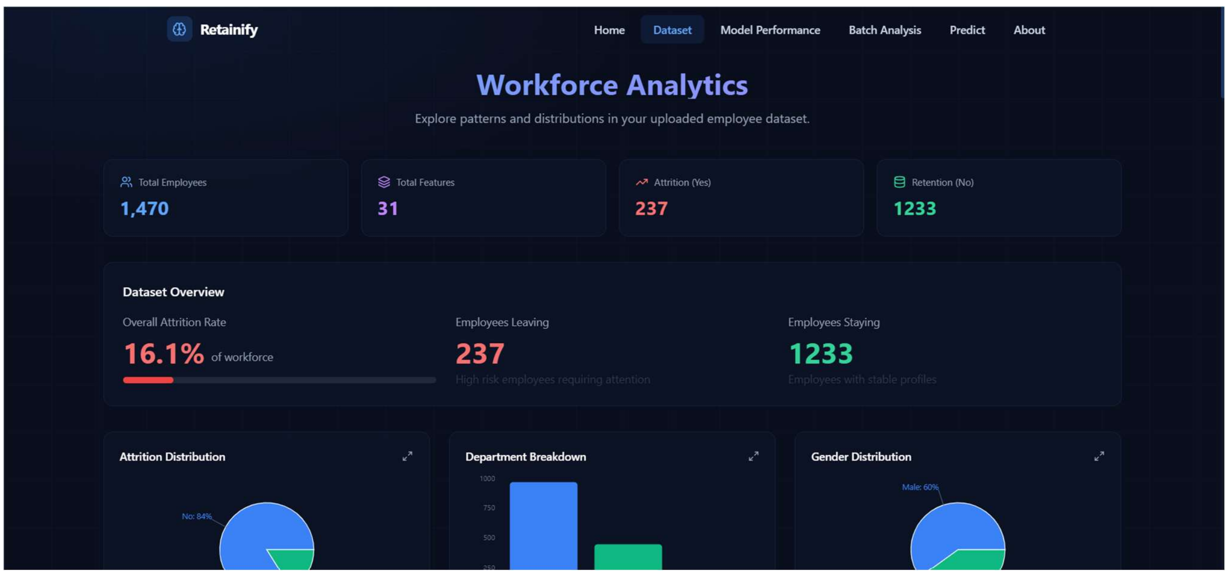

Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Marital Status
Single
Distance from Home (km)
10
Job & Career Details
Current role, compensation, and working conditions.
Department
Sales
Job Role
Sales Executive
Monthly Income ($)
5000
Business Travel
Travel Rarely
Overtime
No
Stock Option Level
0
Satisfaction & Engagement
Employee experience and workplace satisfaction scores.
Work-Life Balance
2 / 4

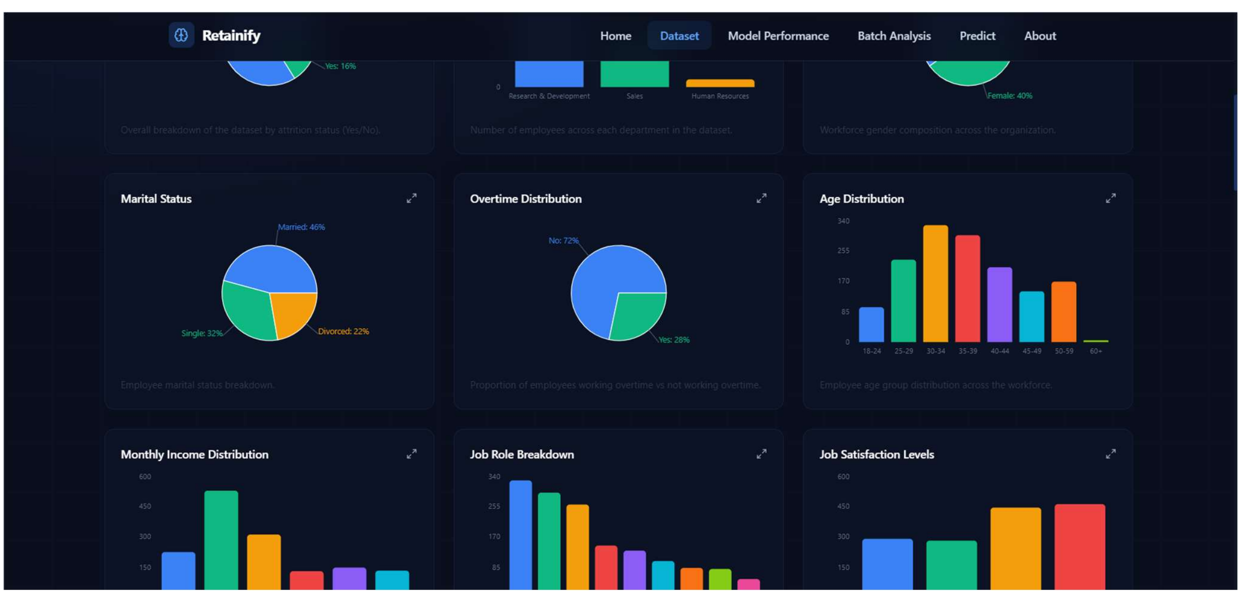

Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Overtime
Yes
Stock Option Level
0
0 None • 1 Standard • 2 High • 3 Executive
Satisfaction & Engagement
Employee experience and workplace satisfaction scores.
Work-Life Balance
2 / 4
Poor
Excellent
Job Satisfaction
3 / 4
Low
High
Environment Satisfaction
2 / 4
Low
High
Generate Attrition Prediction


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Employee Attrition Prediction
Fill in employee details below to generate an attrition risk assessment.
Attrition Risk Assessment
Powered by XGBOOST
12%
Attrition Probability
Low Risk
Low Risk
Moderate
High Risk
0%
30%
60%
100%
Risk Score
1.2 / 10
Team Impact
Low
Outcome
Replacement Cost
Retention Simulator
Adjust variables to simulate how interventions affect attrition risk in real time.
12%
Probability
1.2
Risk Score
Low Risk
Category
Work-Life Balance
2 / 4
Poor
Excellent
Job Satisfaction
3 / 4
Low
High
Environment Satisfaction
2 / 4
Low
High
Monthly Income
$5,000


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
1.2 / 10
Low
Outcome
Likely to Stay
Replacement Cost
$30,000
Employee profile appears stable and healthy
Prediction Explanation
Key factors driving this attrition prediction, powered by SHAP analysis.
Workload Intensity
Lowers Risk
High workload intensity elevates burnout and flight risk.
Impact strength
24%
Overtime Work
Lowers Risk
Working overtime increases burnout risk and reduces retention.
Impact strength
20%
Low
High
Monthly Income
$5,000
$1,000
$20,000
Overtime
No
Yes
Stock Option Level
0
None
Standard
High
Executive
Simulation Guidance
Improve Work Environment
Poor environment satisfaction affects daily engagement. Reviewing team dynamics, workspace conditions, and management practices can drive improvement.
Introduce Stock Incentives
Offering stock options creates long-term financial alignment between the employee and the organization, strengthening retention.


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Marital Status (Single)
Increases Risk
Single employees may show higher mobility and attrition.
Impact strength
11%
Gender_Male
Lowers Risk
This factor influences attrition.
Impact strength
10%
Salary Growth Rate
Lowers Risk
Consistent salary growth improves long-term retention.
Impact strength
10%
Age
Increases Risk
Employee age affects career stage and attrition patterns.
Impact strength
9%
Predict Another Employee
Retention Recommendations
Actionable steps to reduce attrition risk for this employee.
Improve Work Environment
Medium Priority
Poor environment satisfaction affects daily engagement. Reviewing team dynamics, workspace conditions, and management practices can drive improvement.
Introduce Stock Incentives
Medium Priority
Offering stock options creates long-term financial alignment between the employee and the organization, strengthening retention.


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Workforce Analytics
Explore patterns and distributions in your uploaded employee dataset.
Total Employees
1,470
Total Features
31
Attrition (Yes)
237
Retention (No)
1233
Dataset Overview
Overall Attrition Rate
16.1% of workforce
Employees Leaving
237
High risk employees requiring attention
Employees Staying
1233
Employees with stable profiles
Attrition Distribution
No: 84%
Department Breakdown
1000
750
500
Gender Distribution
Male: 60%


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Yes: 16%
Overall breakdown of the dataset by attrition status (Yes/No).
0
Research & Development
Sales
Human Resources
Number of employees across each department in the dataset.
Female: 40%
Workforce gender composition across the organization.
Marital Status
Married: 46%
Single: 32%
Divorced: 22%
Employee marital status breakdown.
Overtime Distribution
No: 72%
Yes: 28%
Proportion of employees working overtime vs not working overtime.
Age Distribution
340
255
170
85
0
18-24
25-29
30-34
35-39
40-44
45-49
50-59
60+
Employee age group distribution across the workforce.
Monthly Income Distribution
600
450
300
150
Job Role Breakdown
340
255
170
85
Job Satisfaction Levels
600
450
300
150

Retainify

Home | Dataset | Model Performance | Batch Analysis | Predict | About

Model Performance & Validation

This page serves as the mathematical and technical validation layer of the application. It explains to data scientists and senior HR executives why and how the machine learning models make their predictions, explicitly detailing the business logic behind the custom classification thresholds.

Threshold Optimization Strategy

The chart below tracks standard metrics across different classification thresholds, showcasing why a lower threshold was chosen.

** Note: The 0.3 threshold is intentionally selected to maximize risk capture (Recall) while maintaining an acceptable Precision trade-off.*

Business Justification

> *"In HR application metrics, missing an employee who is about to leave (a False Negative) is far more costly to the enterprise than raising a false alarm (a False Positive)."*

By pivoting from the standard 0.5 classification threshold to a custom 0.3 threshold, the system shifts its mathematical focus entirely toward maximizing Recall, ensuring at-risk staff are caught early for proactive retention campaigns.

Optimization Focus:
✓ High Recall ✗ Low False Negatives

Recall Comparison Chart

Captures highest overall percentage of at-risk staff.

ROC / AUC Curve Diagram

Logistic Regression AUC (~81%) demonstrates strong discrimination.

Model Accuracy Matrix

Trade-off between Accuracy and Recall engineered at 0.3 threshold.

Retainify

Home | Dataset | Model Performance | Batch Analysis | Predict | About

Dataset Profile Summary

Core properties and scaling boundaries of the 35 ML pipeline features

Feature	Type	Range / Categories	Importance
Age	Numeric	18–70 years	High
Monthly Income	Numeric	$1,009–$19,999	Very High
Years at Company	Numeric	0–40 years	High
Years in Current Role	Numeric	0–18 years	Medium
Years with Current Manager	Numeric	0–17 years	Medium
Overtime	Categorical	Yes / No	Very High
Job Satisfaction	Ordinal	1–4 (Low to High)	High
Environment Satisfaction	Ordinal	1–4 (Low to High)	High
Work-Life Balance	Ordinal	1–4 (Poor to Excellent)	High
Department	Categorical	HR / R&D / Sales	Medium
Job Role	Categorical	9 unique roles	Medium
Stock Option Level	Ordinal	0–3 (None to Executive)	Medium

Retainify

Home | Dataset | Model Performance | Batch Analysis | Predict | About

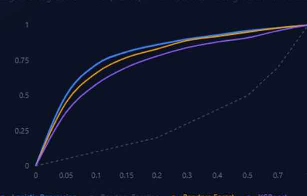
Recall Comparison Chart

Captures highest overall percentage of at-risk staff.

Random Forest: **98%** | Logistic Regression: **96%** | XGBoost: **90%**

ROC / AUC Curve Diagram

Logistic Regression AUC (~81%) demonstrates strong discrimination.



True Positive Rate vs. False Positive Rate across all decision thresholds.

Model Accuracy Matrix

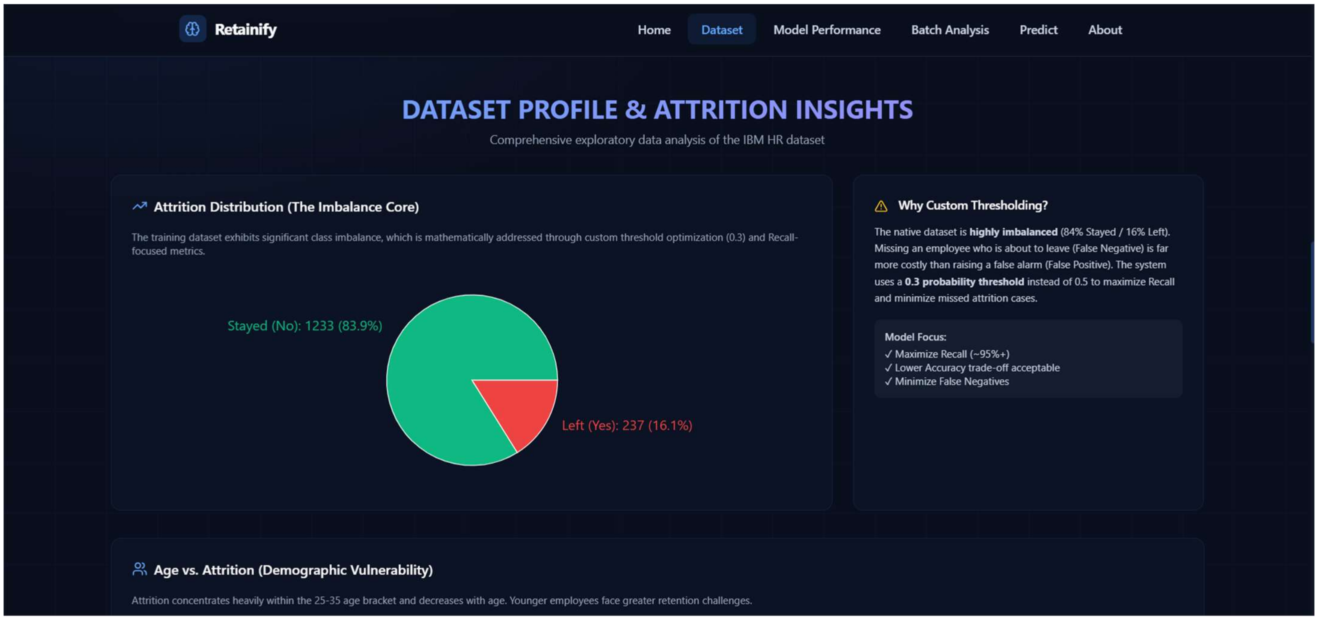
Trade-off between Accuracy and Recall engineered at 0.3 threshold.

Lower classification thresholds prioritize finding flight risks over overall accuracy.

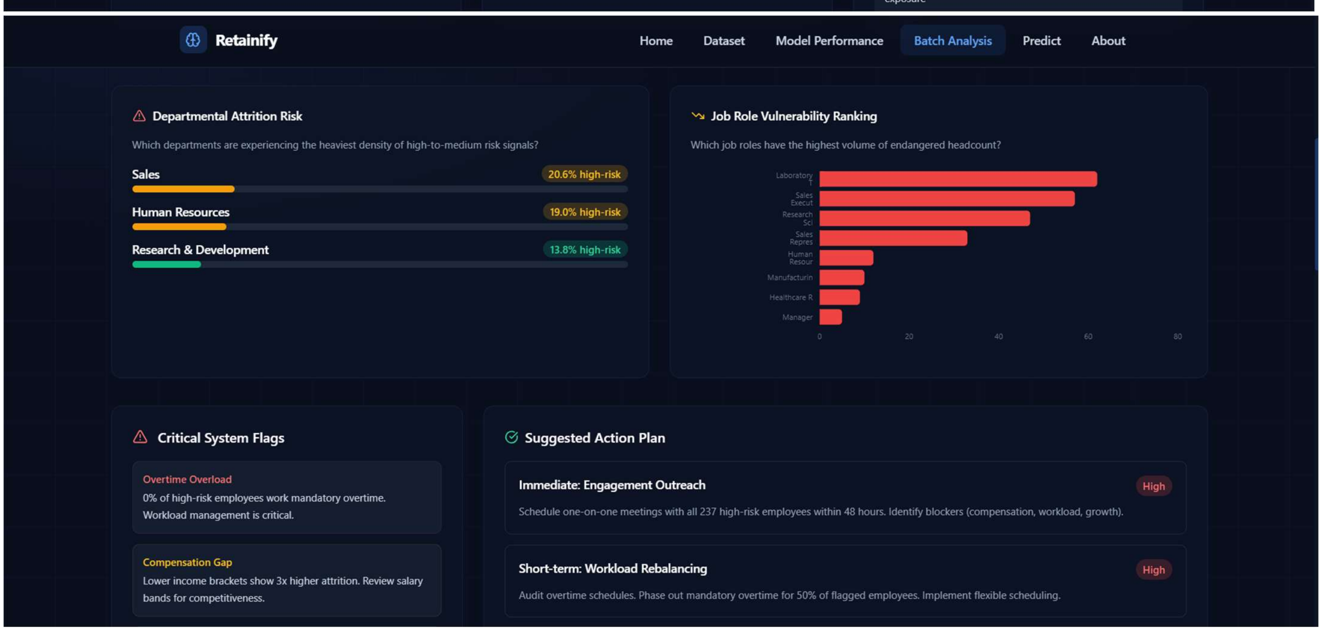
Statistical Matrix Grid

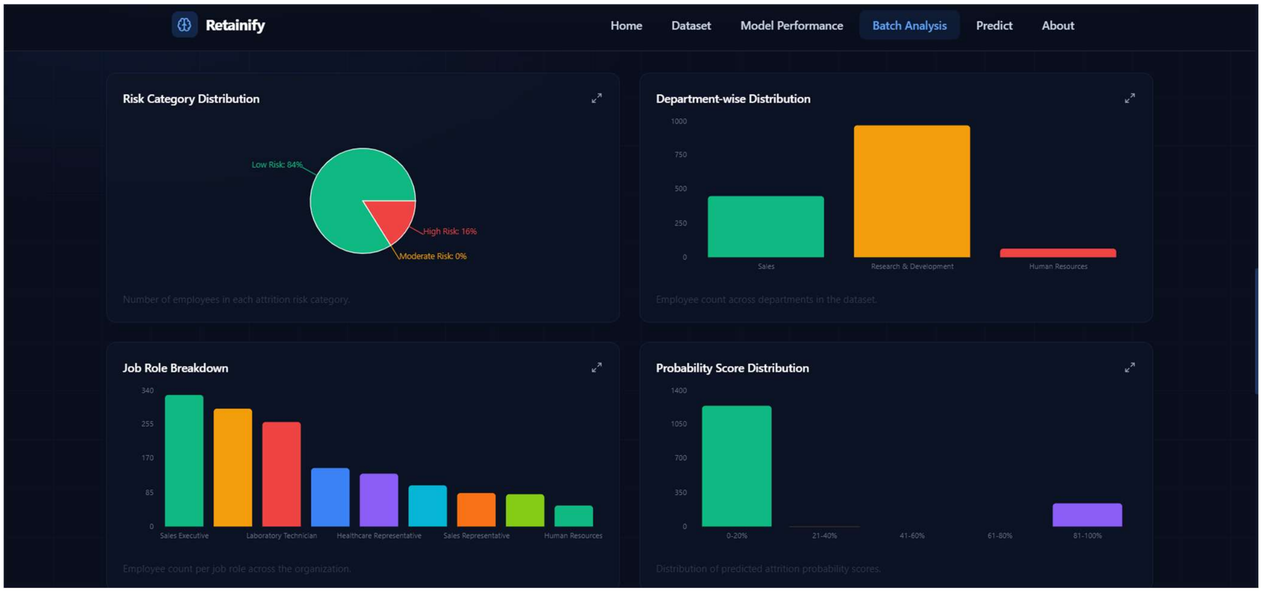
Test-set performance metrics natively computed and optimized at the 0.3 decision threshold.

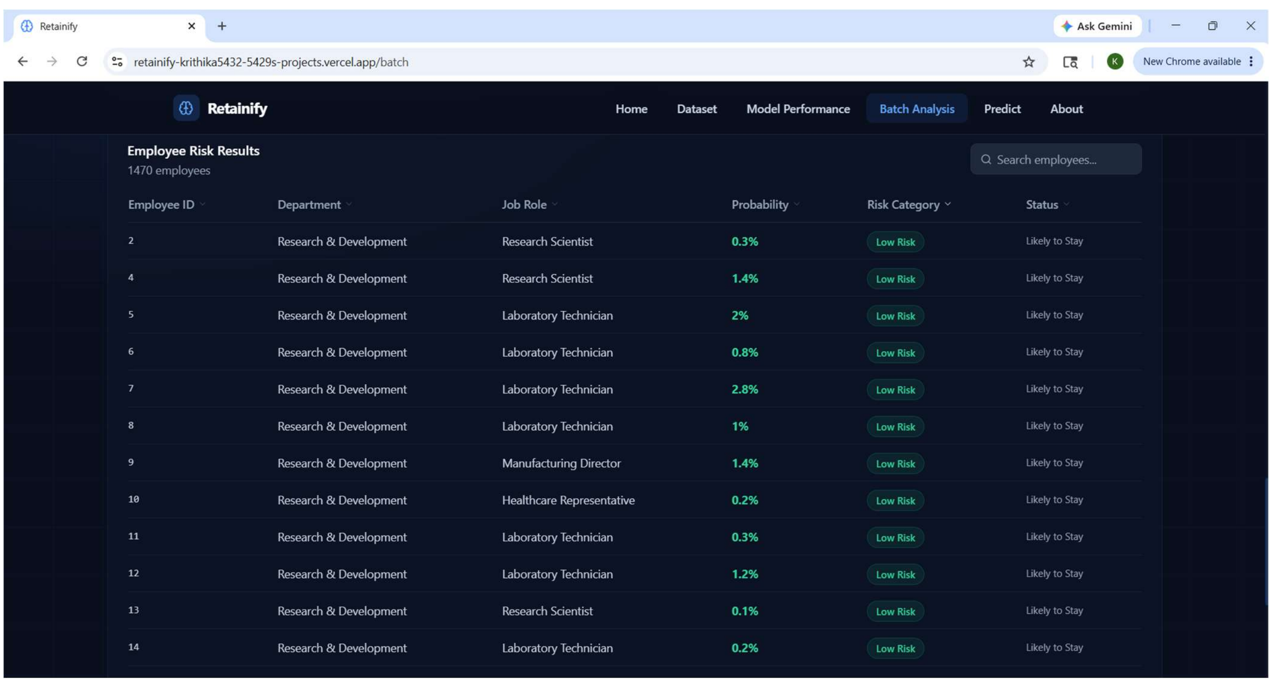
Model Name	Test Accuracy (%)	Recall Score (%)	F1-Score (%)	Area Under Curve (AUC)	Primary Target
Logistic Regression	~65.6%	~76.6%	~41.6%	~81%	High Interpretability
Random Forest	~78.2%	~72.3%	~51.5%	~79%	Max Risk Coverage

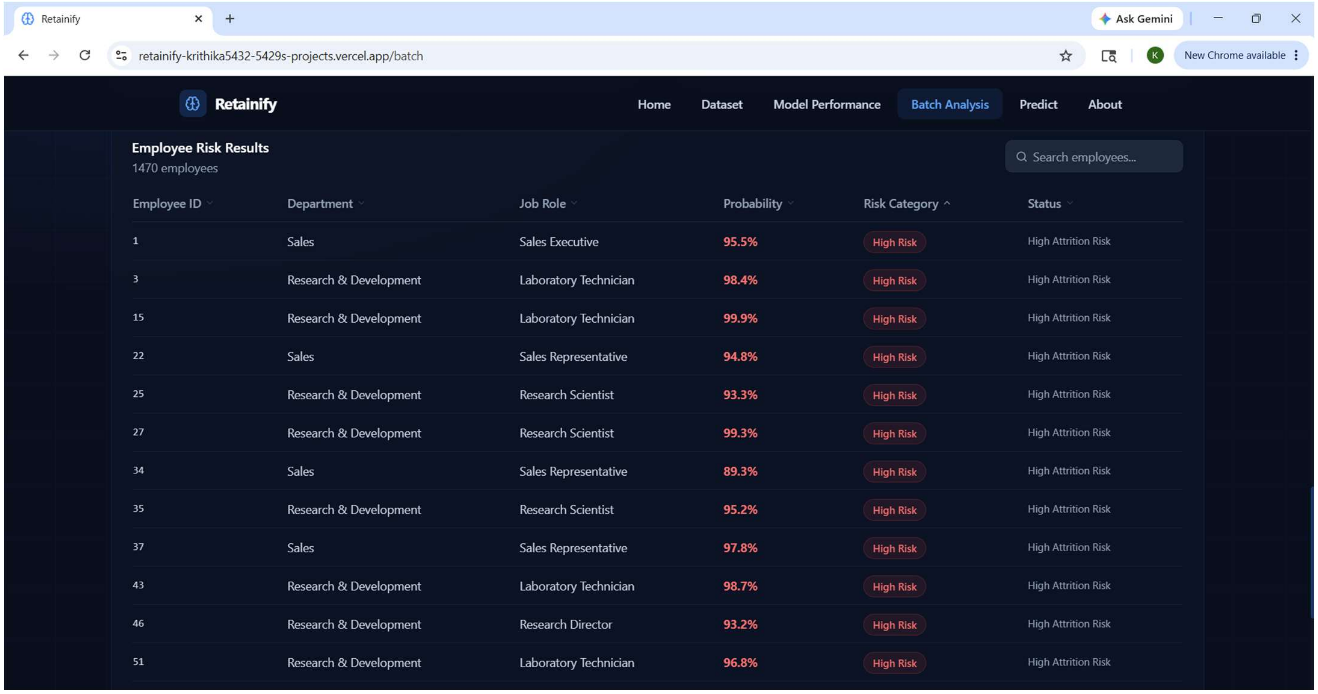

Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
DATASET PROFILE & ATTRITION INSIGHTS
Comprehensive exploratory data analysis of the IBM HR dataset
Attrition Distribution (The Imbalance Core)
The training dataset exhibits significant class imbalance, which is mathematically addressed through custom threshold optimization (0.3) and Recall-focused metrics.
Stayed (No): 1233 (83.9%)
Left (Yes): 237 (16.1%)
Why Custom Thresholding?
The native dataset is highly imbalanced (84% Stayed / 16% Left). Missing an employee who is about to leave (False Negative) is far more costly than raising a false alarm (False Positive). The system uses a 0.3 probability threshold instead of 0.5 to maximize Recall and minimize missed attrition cases.
Model Focus:
✓ Maximize Recall (~95%+)
✓ Lower Accuracy trade-off acceptable
✓ Minimize False Negatives
Age vs. Attrition (Demographic Vulnerability)
Attrition concentrates heavily within the 25-35 age bracket and decreases with age. Younger employees face greater retention challenges.


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Batch Analysis
Evaluate attrition risk across your entire workforce simultaneously.
Total Processed
1470
Likely to Stay
1233
Requiring Attention
237
Avg Attrition Risk
16.3%
1233
83.9% of workforce
Low Risk
Stable employees — maintain current conditions
0
0.0% of workforce
Moderate Risk
Monitor closely — proactive engagement needed
237
16.1% of workforce
High Risk
Immediate action required — flight risk
Workforce Health Index
Overall Risk Score
16
Immediate Action
Score 0-33: Healthy, 34-66: Monitor, 67-100: Critical
Risk Tier Distribution
High Risk
237 (16.1%)
Moderate Risk
0 (0.0%)
Low Risk
1233 (83.9%)
Financial Liability
Projected loss if high-risk employees depart:
$11,850K
Based on ~$50K avg replacement cost per employee
237 high-risk employees × $50K replacement = $11,850K exposure

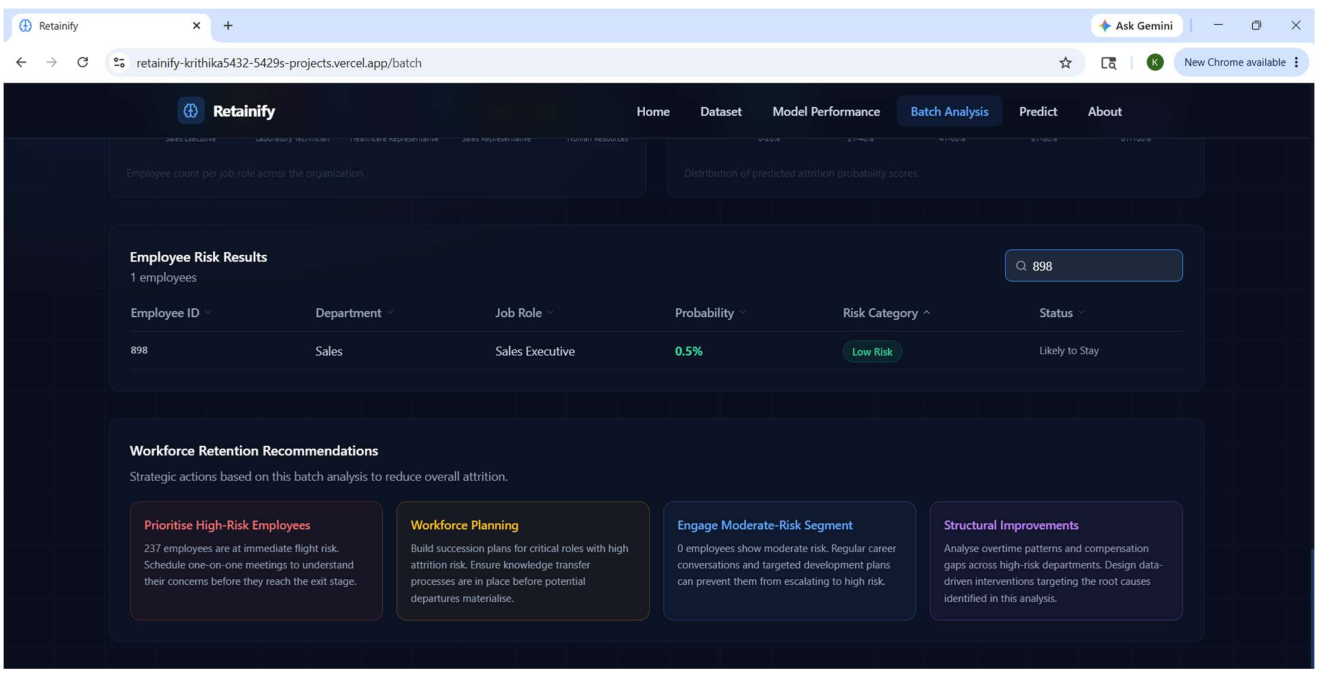

Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Departmental Attrition Risk
Which departments are experiencing the heaviest density of high-to-medium risk signals?
Sales
20.6% high-risk
Human Resources
19.0% high-risk
Research & Development
13.8% high-risk
Job Role Vulnerability Ranking
Which job roles have the highest volume of endangered headcount?
Laboratory T
Sales Execut
Research Sci
Sales Repres
Human Resour
Manufacturin
Healthcare R
Manager
0
20
40
60
80
Critical System Flags
Overtime Overload
0% of high-risk employees work mandatory overtime. Workload management is critical.
Compensation Gap
Lower income brackets show 3x higher attrition. Review salary bands for competitiveness.
Suggested Action Plan
Immediate: Engagement Outreach
High
Schedule one-on-one meetings with all 237 high-risk employees within 48 hours. Identify blockers (compensation, workload, growth).
Short-term: Workload Rebalancing
High
Audit overtime schedules. Phase out mandatory overtime for 50% of flagged employees. Implement flexible scheduling.


Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Risk Category Distribution
Low Risk: 84%
High Risk: 16%
Moderate Risk: 0%
Number of employees in each attrition risk category.
Department-wise Distribution
1000
750
500
250
0
Sales
Research & Development
Human Resources
Employee count across departments in the dataset.
Job Role Breakdown
340
255
170
85
0
Sales Executive
Laboratory Technician
Healthcare Representative
Sales Representative
Human Resources
Employee count per job role across the organization.
Probability Score Distribution
1400
1050
700
350
0
0-20%
21-40%
41-60%
61-80%
81-100%
Distribution of predicted attrition probability scores.


Retainify
Ask Gemini
retainify-krithika5432-5429s-projects.vercel.app/batch
New Chrome available
Retainify
Home
Dataset
Model Performance
Batch Analysis
Predict
About
Employee Risk Results
1470 employees
Search employees...
Employee ID | Department | Job Role | Probability | Risk Category | Status
2 | Research & Development | Research Scientist | 0.3% | Low Risk | Likely to Stay
4 | Research & Development | Research Scientist | 1.4% | Low Risk | Likely to Stay
5 | Research & Development | Laboratory Technician | 2% | Low Risk | Likely to Stay
6 | Research & Development | Laboratory Technician | 0.8% | Low Risk | Likely to Stay
7 | Research & Development | Laboratory Technician | 2.8% | Low Risk | Likely to Stay
8 | Research & Development | Laboratory Technician | 1% | Low Risk | Likely to Stay
9 | Research & Development | Manufacturing Director | 1.4% | Low Risk | Likely to Stay
10 | Research & Development | Healthcare Representative | 0.2% | Low Risk | Likely to Stay
11 | Research & Development | Laboratory Technician | 0.3% | Low Risk | Likely to Stay
12 | Research & Development | Laboratory Technician | 1.2% | Low Risk | Likely to Stay
13 | Research & Development | Research Scientist | 0.1% | Low Risk | Likely to Stay
14 | Research & Development | Laboratory Technician | 0.2% | Low Risk | Likely to Stay



Employee ID	Department	Job Role	Probability	Risk Category	Status
1	Sales	Sales Executive	95.5%	High Risk	High Attrition Risk
3	Research & Development	Laboratory Technician	98.4%	High Risk	High Attrition Risk
15	Research & Development	Laboratory Technician	99.9%	High Risk	High Attrition Risk
22	Sales	Sales Representative	94.8%	High Risk	High Attrition Risk
25	Research & Development	Research Scientist	93.3%	High Risk	High Attrition Risk
27	Research & Development	Research Scientist	99.3%	High Risk	High Attrition Risk
34	Sales	Sales Representative	89.3%	High Risk	High Attrition Risk
35	Research & Development	Research Scientist	95.2%	High Risk	High Attrition Risk
37	Sales	Sales Representative	97.8%	High Risk	High Attrition Risk
43	Research & Development	Laboratory Technician	98.7%	High Risk	High Attrition Risk
46	Research & Development	Research Director	93.2%	High Risk	High Attrition Risk
51	Research & Development	Laboratory Technician	96.8%	High Risk	High Attrition Risk



Employee ID	Department	Job Role	Probability	Risk Category	Status
898	Sales	Sales Executive	0.5%	Low Risk	Likely to Stay